

Discrete Mathematics: Algorithms revision

1 Consider the following algorithm, which takes positive integers as inputs:

Step 1: Input A
Step 2: Input B
Step 3: If $A > B$ then let $A = A - B$
Step 4: If $B > A$ then let $B = B - A$
Step 5: If $A = B$ then Output A . STOP
Step 6: Go to Step 3

- (a) Trace through the algorithm with inputs $A = 68$ and $B = 52$. Show the values of A and B each time Step 6 is encountered, and give the final output.
- (b) How many times is Step 6 encountered when the inputs are $A = 60$ and $B = 40$?
- (c) What is the maximum number of times that Step 6 could be encountered?
- (d) Explain why the instructions above would not comprise an algorithm if the requirement that the inputs are positive were removed.

2 The table below shows the sizes of 10 files. These files need to be put onto USB sticks with a maximum capacity of 16 GB.

File	A	B	C	D	E	F	G	H	I	J
Size (GB)	12.4	8.1	2.3	1.2	7.3	7.4	1.7	8.8	7.5	3.2

Show the packing that is produced when the following algorithms are used:

- (a) next-fit
- (b) first-fit
- (c) first-fit decreasing.

3 Use the shuttle sort to sort the list 6, 4, 8, 2, 4, 13, 5 into ascending order. Show the list at the end of each pass of the algorithm, and record the numbers of comparisons and swaps involved in each pass of the algorithm.

4 Two different algorithms are being used to process a set of n objects. Algorithm A has order n^2 and algorithm B has order $n \log n$. When Algorithm A is used on a set of 500 objects it takes 2.4 seconds to complete. When Algorithm B is used on the same set of 500 objects it takes 10.7 seconds to complete. Roughly how long will each algorithm take when used on a set of:

- (a) 2000 objects?
- (b) 10,000 objects?

- 5 Tamsin is sorting a list of numbers into increasing order. After the second pass of her algorithm she has the following list:

8 6 1 10 5 6 12

- (a) Explain how you know that Tamsin is not using the Shuttle Sort.

In fact, Tamsin is using the Bubble Sort.

- (b) Find all the possibilities for Tamsin's original list.

- (c) How many comparisons in total are needed to sort Tamsin's list?

- 6 The following 16 items need to be packed into boxes which have maximum capacity of 10 kg.

Item	A	B	C	D	E	F	G	H
Mass (kg)	6.4	1.5	8.1	5.0	2.2	7.4	1.8	8.7
Item	I	J	K	L	M	N	O	P
Mass (kg)	2.4	2.8	3.6	2.5	3.6	6.9	1.9	6.7

- (a) Use the full bins method, followed by the first-fit decreasing algorithm to find a possible packing.

- (b) Explain how you know that the packing you found is optimal.

- 7 Wilfred is sorting a list of numbers into increasing order. After the third pass of his algorithm he has the following list:

2 5 7 10 3 9 12

- (a) Explain how you know that Wilfred is not using a bubble sort.

In fact, Wilfred is using the shuttle sort.

- (b) How many possibilities are there for the Wilfred's original list?

- (c) What are the maximum and minimum numbers of comparisons that Wilfred has done so far?

8 The following instructions operate on positive integers greater than 4.

STEP 10 Choose any positive integer greater than 4, and call it n .

STEP 15 Write down n .

STEP 20 If n is even then let $n = \frac{n}{2}$ and write down the result

STEP 30 If n is odd then let $n = 3n + 1$ and write down the result.

STEP 40 Go to Step 20.

- (a) Apply the instructions with 6 as the chosen integer, stopping when a sequence repeats itself.
- (b) Apply the instructions with 256 as the chosen integer, stopping when a sequence repeats itself.
- (c) Add an instruction to stop the process when n becomes equal to 1.
- (d) It is not known if, when modified as in Part (c), the instructions form an algorithm. What would need to be known for it to be an algorithm?

9 Lists of numbers are created by starting with the numbers 1 to n in order. The number 1 is then swapped at random with one of the numbers 2 to n .

- (a) What is the maximum and minimum number of comparisons that will be needed when sorting lists of this type using the Shuttle sort?
- (b) Hence deduce the average time complexity, in terms of n , of the shuttle sort when applied to lists of this type.

10 Lists of numbers are created by starting with the numbers 1 to n in order. The number 1 is then swapped at random with one of the numbers 2 to n .

- (a) If the number 1 is swapped with the number k , for $k = 2, 3, \dots, n$, then how many passes of the bubble sort algorithm will be required to sort the resulting list? How many comparisons will be required?
- (b) Hence deduce the average time complexity, in terms of n , of the bubble sort when applied to lists of this type.